Compound interest tables

1

a

	Total balance	Interest earned
First year	$\$3900$	$\$390$
Second year	$\$4290$	$\$429$
Third year	$\$4719$	$\$471.9$
Fourth year	$\$5190.90$	$-$

b $\$1290.90$

2

a

	Balance at beginning of year	Interest earned
First year	$\$3700$	$\$259$
Second year	$\$3959$	$\$277.13$
Third year	$\$4236.13$	$\$296.53$
Fourth year	$\$4532.06$	$-$

b $\$832.66$

3

a

i $A = \$3120$ ii $B = \$120$ iii $C = \$3120$ iv $D = \$124.80$

v $E = \$129.79$

b $\$374.59$

4

a

	Interest ($)	Balance ($)
Starting balance	0	9000
After one year	450	9450
After two years	472.50	9922.50
After three years	496.13	$10\,418.63$

b $\$1418.63$

c $\$10\,418.63$

5

a $140
b $1540
c $154
d $1694
e $169.40
f $1863.40
g $463.40
h 33.1%
i $420
j Compound interest is better by $43.40.

6

a $119 828.00
b $69 828.00

7 12 years

Compound interest formula

8

a $2524.95
b $1061.52
c $14 503.07
d $5305.20
e $11 494.18
f $13 593.90
g $1951.24
h $3757.38
i $547.84

9

a Plan 1: $1219.92

Plan 2: $1291.17

Plan 3: $1248.25

b Plan 2

10 $611.65

11 $31.09

12 $14 929.92

13 $8034.12

Calculate the principal

14 $1338.79

16 $5581.03

17

a $7867.54
b $7823.90
c $7801.70
d $7786.76
e $7780.98
f $7779.49

18 $49 909.03